Corollary

If in ΔABC , a straight line DE parallel to BC, intersects AB at D and AC at E, then

(i) $\frac{AB}{AD} = \frac{AC}{AE}$ (ii) $\frac{AB}{DB} = \frac{AC}{EC}$.

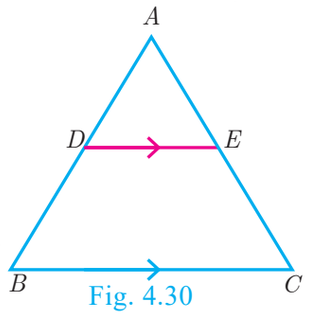


Fig. 4.30

Proof

In $\Delta ABC,\ DE \parallel BC$,

Therefore, $\frac{AD}{DB} = \frac{AE}{EC}$ (by Basic Proportionality Theorem)

(i) Taking reciprocals, we get $\frac{DB}{AD} = \frac{EC}{AE}$

Add 1 to both in the sides $\frac{DB}{AD} + 1 = \frac{EC}{AE} + 1$

$\frac{DB + AD}{AD} = \frac{EC + AE}{AE}$ so, $\frac{AB}{AD} = \frac{AC}{AE}$

(ii) Add 1 to both the sides

$\frac{AD}{DB} + 1 = \frac{AE}{EC} + 1$

Therefore, $\frac{AB}{DB} = \frac{AC}{EC}$

Is the converse of Basic Proportionality Theorem also true? To examine let us do the following illustration.

Illustration

Draw an angle XAY on your notebook as shown in Fig.4.31 and on ray AX, mark points B_1, B_2, B_3, B_4 and B such that $AB_1 = B_1B_2 = B_2B_3 = B_3B_4 = B_4B = 1$ cm.

Similarly on ray AY, mark points C_1, C_2, C_3, C_4 and C, such that

$AC_1 = C_1C_2 = C_2C_3 = C_3C_4 = C_4C = 2$ cm, Join B_1C_1 and BC.

Observe that $\frac{AB_1}{B_1B} = \frac{AC_1}{C_1C} = \frac{1}{4}$ and $B_1C_1 \parallel BC$

Similarly joining B_2C_2, B_3C_3 and B_4C_4 you see that

$\frac{AB_2}{B_2B} = \frac{AC_2}{C_2C} = \frac{2}{3}$ and $B_2C_2 \parallel BC$

$\frac{AB_3}{B_3B} = \frac{AC_3}{C_3C} = \frac{3}{2}$ and $B_3C_3 \parallel BC$

$\frac{AB_4}{B_4B} = \frac{AC_4}{C_4C} = \frac{4}{1}$ and $B_4C_4 \parallel BC$

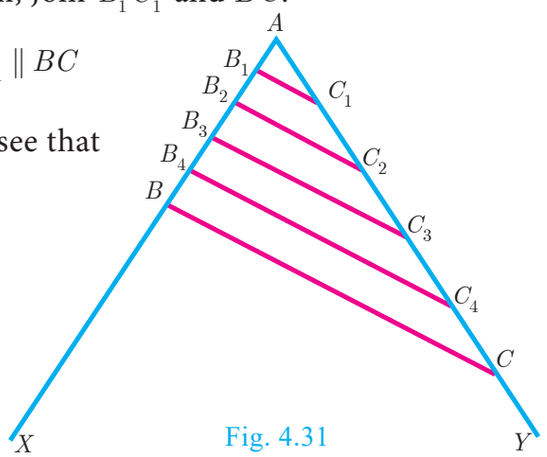


Fig. 4.31

From this we observe that if a line divides two sides of a triangle in the same ratio, then the line is parallel to the third side.

Therefore, we obtain the following theorem called converse of the Thales theorem.

Theorem 2: Converse of Basic Proportionality Theorem

Statement

If a straight line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

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